Predicting Diagnosis Related Groups From ICD-10 Codes: A Case Study at Hospital El Pino

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CINF104 — May 2026

Keywords: diagnosis related group, ICD-10, random forest, class imbalance, hospital coding.

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1 Introduction

Hospitals deal with thousands of patients every year, each with different diagnoses, treatments, and resource needs. To make sense of this complexity, the Diagnosis Related Group (DRG) system groups patients into categories based on their clinical profile—specifically their diagnoses, procedures, age, and sex—and assigns each group a unique six-digit code. In Chile, public hospitals like Hospital El Pino rely on this system to determine how much funding they receive and to measure operational efficiency [1, 2].

Assigning DRG codes manually is tedious and error-prone: it requires navigating thousands of ICD-10 rules, and even small mistakes can lead to incorrect billing or misallocated resources [2]. This raises a natural question: can a machine learning model learn to predict DRG codes automatically from patient data?

Several researchers have explored this direction. Rule-based approaches have shown that DRG assignment can be partially replicated with decision trees [3], while more recent work demonstrates that ensemble methods such as random forest perform well on multi-class clinical coding [4]. Deep learning with medical code embeddings has achieved strong results on large hospital datasets [5]. However, these studies often use large, well-balanced datasets from multiple institutions—conditions rarely available in smaller hospitals.

In this study, we use real data from Hospital El Pino, a public hospital in Santiago. The dataset includes 14,561 patient records with ICD-10 diagnosis codes, procedure codes, age, and sex, spanning 526 unique DRG classes. We train and compare logistic regression and random forest and address severe class imbalance through rare-class filtering and class weighting.

2 Study Objectives

The general objective is to develop and evaluate the prediction capability of the selected machine learning models for the Diagnosis Related Group (DRG) of patients from Hospital

El Pino, based on ICD-10 diagnosis and procedure codes, age, and sex.

Specific objectives:

1. Perform exploratory data analysis (EDA) of the Hospital El Pino dataset: data quality, class distribution, completeness, and relevant clinical patterns.
2. Design and implement a preprocessing pipeline that transforms raw ICD-10 codes into feature vectors, addressing imbalance through rare-class filtering and class weighting.
3. Train and compare logistic regression and random forest using weighted F1-score, macro F1-score, and accuracy, selecting the best model from experimental evidence.

3 Methodology

3.1 Dataset description

The dataset contains 14,561 patient records from hospital stays, with primary diagnosis, secondary diagnoses, primary and secondary procedures, age, sex, and a DRG label (six-digit code representing the diagnosis related group). A salient fact is strong class imbalance: 297 of 526 DRG classes (56.5%) have fewer than ten patient records, while a small number of classes concentrate most cases. One outlier was identified in age (121 years). Beyond the first diagnosis column, missing values appear as the placeholder "-".

3.2 Development process

The pipeline follows four stages.

Stage 1 — Data loading and cleaning. Data are loaded (e.g., with pandas and semicolon delimiter). ICD-10 codes are extracted with a regular expression that isolates the code prefix from the text description. Missing values "-" and NaN are filtered before feature construction.

Stage 2 — Exploratory data analysis. EDA assesses completeness, outliers, and class distribution; computes descriptive statistics; and visualizes age and sex distributions, DRG frequency, number of codes per patient, completeness of diagnosis columns, and the top frequent ICD-10 diagnosis codes, highlighting patterns characteristic of this hospital.

Stage 3 — Feature engineering and class balancing. Diagnosis and procedure codes are encoded separately using `MultiLabelBinarizer`, yielding a sparse binary matrix (each column is a unique code, each row a patient). Age and sex are appended. DRG classes with fewer than ten samples are removed before training, reducing the label space from 526 to 229 classes and retaining 13,454 records. Remaining imbalance is mitigated with `class_weight='balanced'` in training.

Stage 4 — Model training and evaluation. Two models are trained with an 80/20 stratified train–test split. The best model is selected primarily by weighted F1-score.

3.3 Machine learning techniques

Logistic regression is the linear baseline: efficient, interpretable, and suited to high-dimensional sparse inputs. We use the SAGA solver with regularization $C = 0.1$.

Random forest is the main nonlinear model: 100 trees, maximum depth 15, `max_features='sqrt'`, bootstrap aggregating, and variance reduction via random feature subsets at splits.

Both models use `class_weight='balanced'` to up-weight minority classes.

3.4 Evaluation metrics

Given multi-class prediction and imbalance, accuracy alone is insufficient (a majority-class classifier can look deceptively good). We report **weighted F1** (emphasizes frequent DRG groups) and **macro F1** (equal weight per class). A large gap between them indicates poor performance on rare classes.

4 Exploratory Data Analysis

4.1 Data quality

Completeness: Diag01 is 100% complete across 14,561 records; completeness decreases with each secondary diagnosis column, reflecting sparse secondary coding. Procedures follow a similar pattern. Missing cells use "-" and are treated as absent codes rather than imputed.

Correctness: Primary diagnosis codes match the expected ICD-10 pattern; sex takes values *Mujer* / *Hombre*; the GRD column consistently shows a six-digit prefix plus description.

Outliers: One age value is 121 years; no negative ages. The age distribution is right-skewed (median 36 years), consistent with obstetric and pediatric case mix.

4.2 Descriptive statistics

Figure 1 reproduces the descriptive statistics screenshot. Table 1 lists the same numbers in tabular form (including the maximum column). Female patients: 9,617 (66%); male: 4,944 (34%). Among 526 DRG classes, 297 (56.5%) have fewer than ten records each, jointly only 1,107 patients (7.6%).

Variable	Count	Mean	Std	Min	25%	Median	75%	Max
Age (years)	14,561	39.4	24.7	0	23	36	60	121
Diagnoses per patient	14,561	3.8	3.1	1	1	3	6	35
Procedures per patient	14,561	4.2	5.6	0	0	2	7	30

Figure 1: Descriptive statistics of numeric variables (count, mean, std, quartiles, min/max).

Table 1: Descriptive statistics of numeric constructs (same values as Fig. 1).

Variable	Count	Mean	Std	Min	25%	Median	75%	Max
Age (years)	14,561	39.4	24.7	0	23	36	60	121
Diagnoses per patient	14,561	3.8	3.1	1	1	3	6	35
Procedures per patient	14,561	4.2	5.6	0	0	2	7	30

4.3 Visualizations

We provide overview plots in Figs. 2–4. The age histogram shows a right-skewed distribution with concentration between 20–40 years and a spike at age 0 (neonatal/obstetric mix), plus a long right tail including the 121-year outlier. Completeness of diagnosis columns drops rapidly after the sixth secondary field, supporting list-based encoding. The top ICD-10 codes include I10, Z39.x, Z38.0, Z37.0, consistent with obstetric episodes. The bottom-right panel of Fig. 2 shows DRG class counts on a log scale (long tail, severe imbalance).

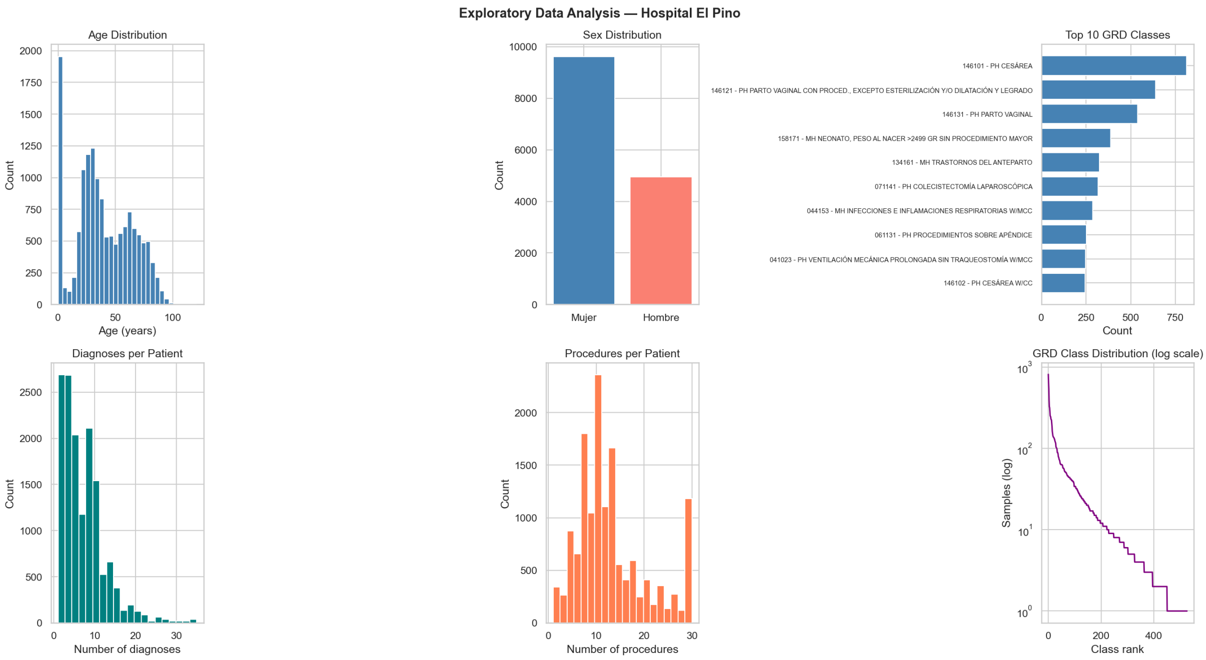

Figure 2: Exploratory overview: age, sex, top DRGs, counts of diagnoses/procedures per patient, and DRG class frequency (log scale, bottom right).

5 Experiments

5.1 Feature and target selection

Inputs: ICD-10 diagnosis codes, procedure codes, age, and sex. Text descriptions are omitted to avoid redundant high-cardinality features. The target is the six-digit DRG code from the GRD column. After filtering rare classes, 229 DRG labels remain.

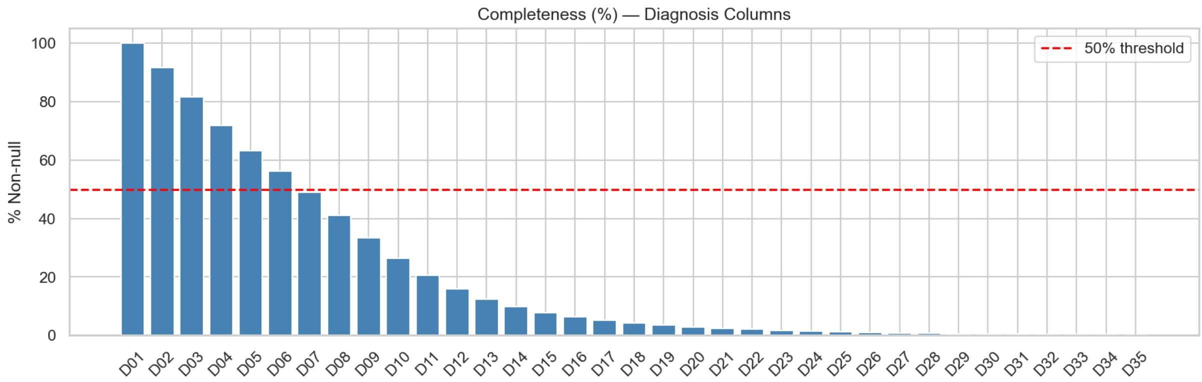

Figure 3: Completeness (% non-null) of secondary diagnosis columns (Diag02–Diag35).

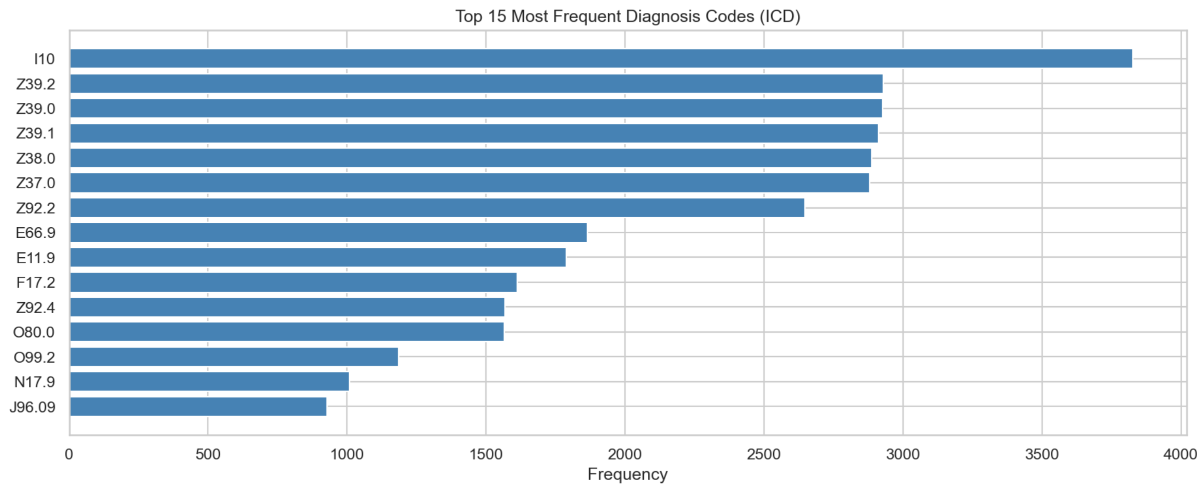

Figure 4: Fifteen most frequent ICD-10 diagnosis codes.

5.2 Model training results

Training: 10,763 samples; test: 2,691 (stratified split). Figure 5 and Table 2 summarize metrics. Random forest outperforms logistic regression on all three reported metrics; weighted F1 improves by 10.8 percentage points (0.5107 vs. 0.4026). Random forest is selected as the final model.

Model	Accuracy	F1 Weighted	F1 Macro
Logistic Regression	0.4051	0.4026	0.2699
Random Forest	0.5206	0.5107	0.3161

Figure 5: Test-set comparison: logistic regression vs. random forest (accuracy, weighted F1, macro F1).

Table 2: Test-set metrics (same values as Fig. 5).

Model	Accuracy	F1 (weighted)	F1 (macro)
Logistic regression	0.4051	0.4026	0.2699
Random forest	**0.5206**	**0.5107**	**0.3161**

5.3 Model architecture and training

Figure 6 sketches the end-to-end pipeline. Figure 7 illustrates random forest: patient features, parallel trees, majority vote, predicted DRG.

Random forest uses 100 trees trained on bootstrap samples; each split considers a random feature subset (`max_features='sqrt'`); max depth is 15; `class_weight='balanced'` reweights by class frequency. Input dimension: 4,168 sparse binary features (3,342 diagnosis + 824 procedure + age + sex). Output: one of 229 DRG classes. Training completed without early stopping; wall time is on the order of minutes on a standard workstation.

5.4 Training process visualization

Figure 8 shows out-of-bag error vs. number of trees (5–100). Error decreases sharply in the first ∼50 trees and flattens after ∼80, supporting 100 estimators. Final OOB error ≈ 0.56 is consistent with test accuracy ≈ 0.52, suggesting limited overfitting. Figure 9 shows the same model comparison as a bar chart.

5.5 Performance analysis

Figure 10 lists per-class test metrics. Classes with F1 > 0.85 often belong to obstetric/neonatal DRGs (146xxx) with large support. Class 044163 has low recall (ambiguous overlap with neighboring DRGs, fewer examples). Weighted F1 (≈ 0.51) vs. macro F1 (≈ 0.32) indicates stronger performance on frequent than on rare classes. Table 3 extracts the same rows and the weighted-average line.

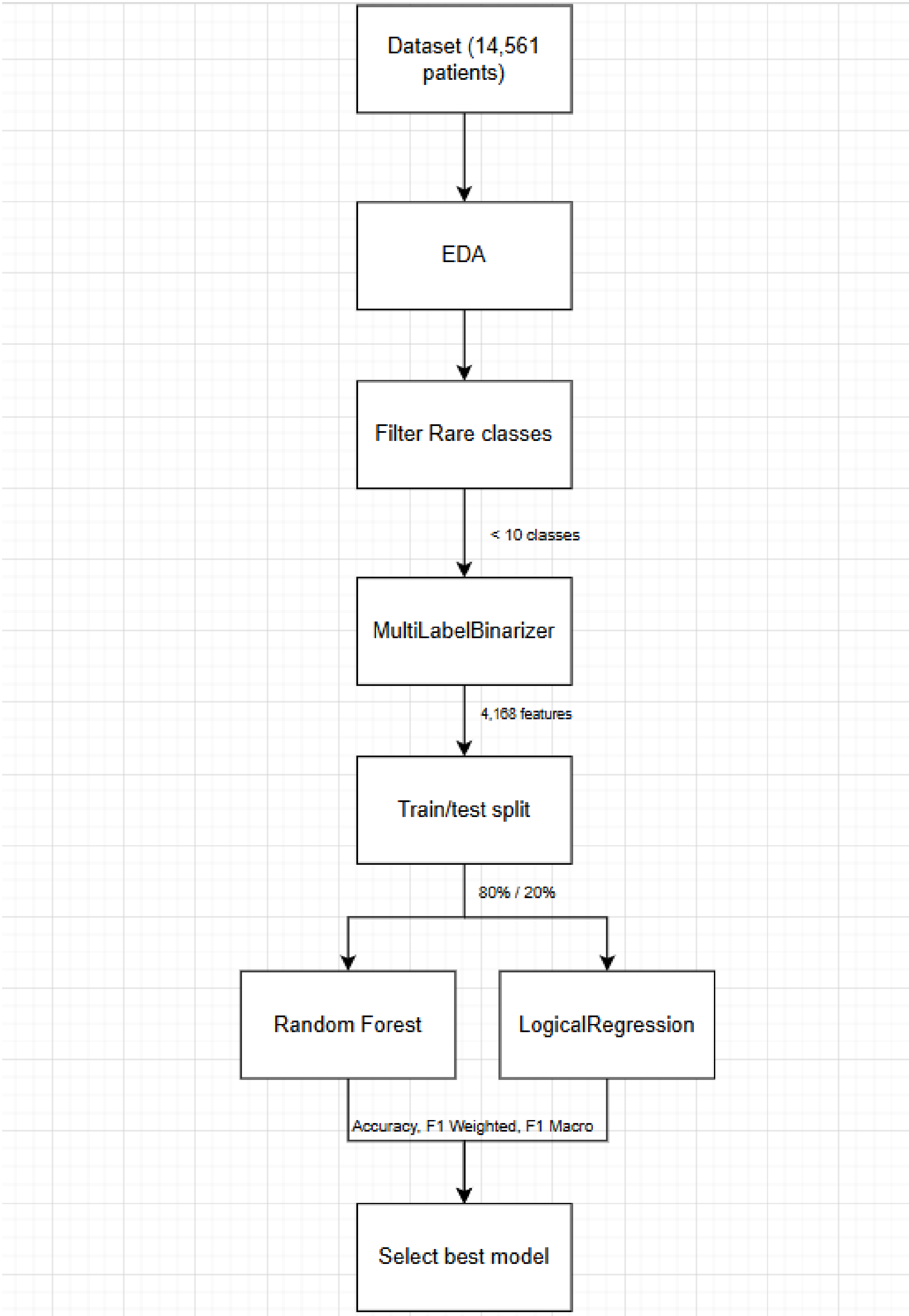

Figure 6: End-to-end experimental pipeline: dataset, EDA, rare-class filter, encoding, train/test split, models, and selection by accuracy / F1 metrics.

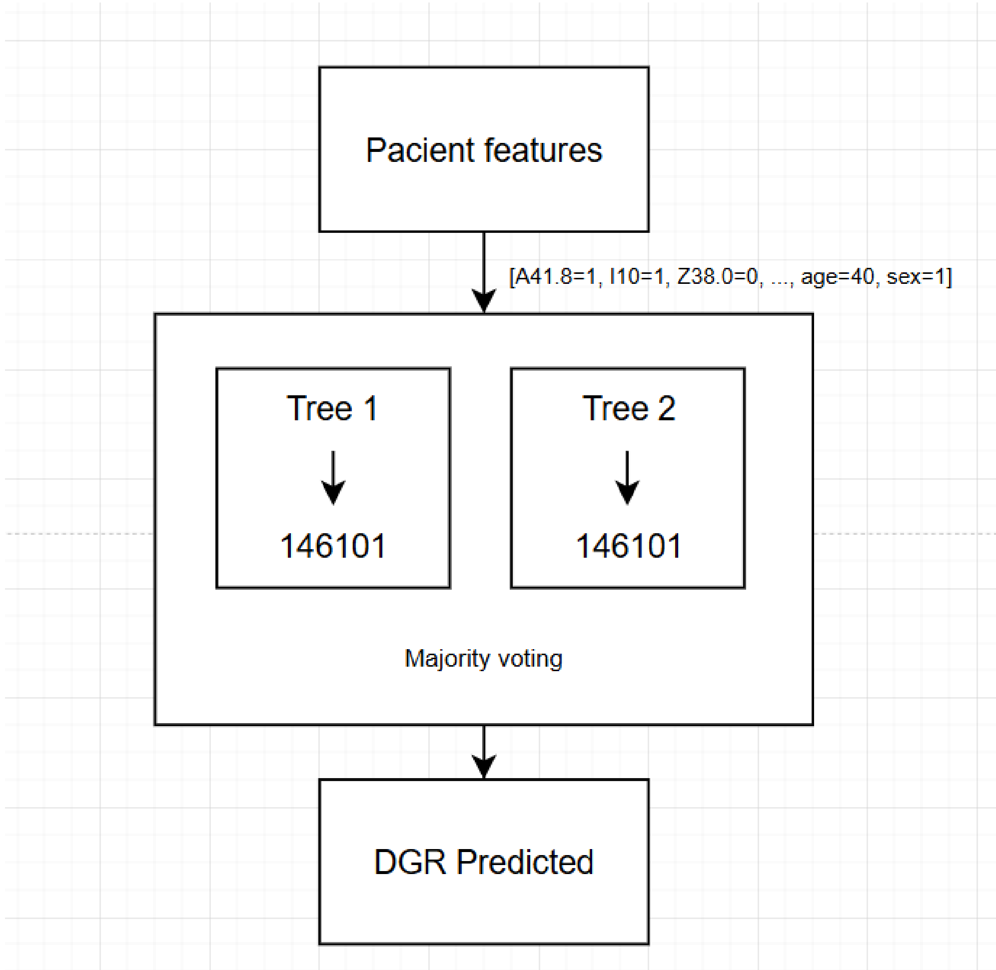

Figure 7: Random forest: sparse patient features, ensemble of decision trees, majority voting, predicted DRG.

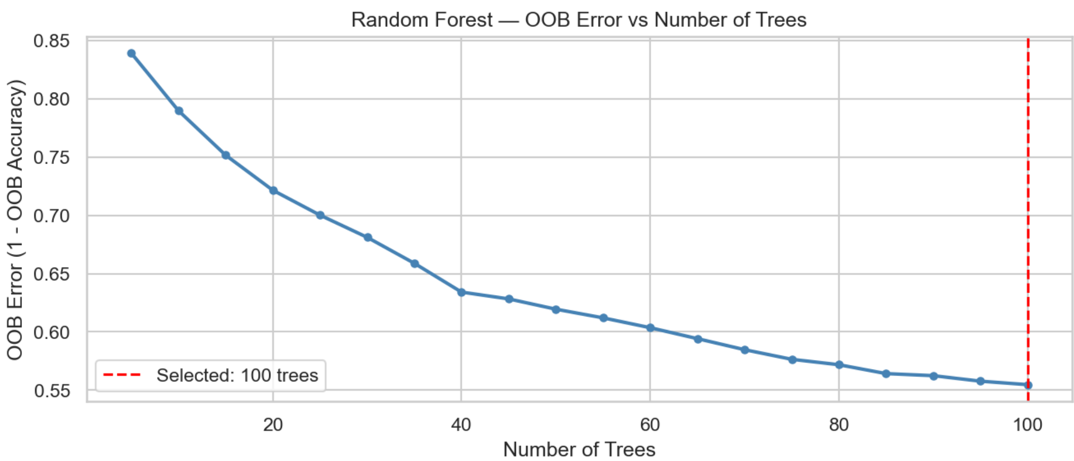

Figure 8: Random forest: out-of-bag error as a function of the number of trees.

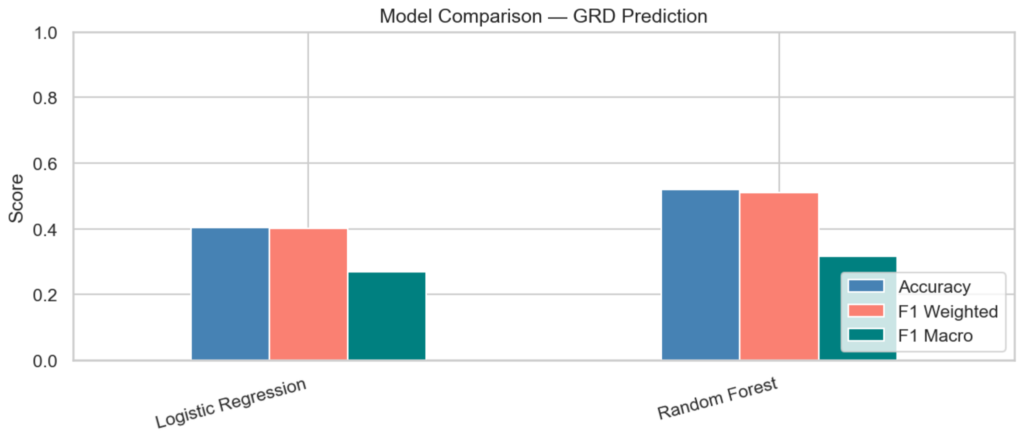

Figure 9: Bar chart comparison of accuracy and F1 metrics across models.

DRG Code	Precision	Recall	F1-Score	Support
146101	0.825	0.896	0.859	163
146121	0.830	0.953	0.887	128
146131	0.856	0.880	0.868	108
061131	0.857	0.941	0.897	51
041023	0.723	0.940	0.817	50
044163	0.400	0.087	0.143	46
Weighted avg	0.544	0.521	0.511	2,691

Figure 10: Per-class test metrics (random forest) and weighted averages; total support = 2,691 test samples.

Table 3: Example per-class test metrics and weighted average (same values as Fig. 10).

DRG	Prec.	Rec.	F1	Supp.
146101	0.825	0.896	0.859	163
146121	0.830	0.953	0.887	128
146131	0.856	0.880	0.868	108
061131	0.857	0.941	0.897	51
041023	0.723	0.940	0.817	50
044163	0.400	0.087	0.143	46
Weighted avg	0.544	0.521	0.511	2,691

5.6 Comparison with related work

Rule-based groupers on large Medicare-style corpora often reach higher reported accuracy than small single-hospital experiments [3]. Benchmarks that use larger multi-site cohorts and heavy feature engineering frequently exceed our weighted F1 of 0.51; the gap is consistent with dataset size, 229 output classes after filtering, and a simple bag-of-code representation rather than embeddings or hierarchical models [5].

6 Conclusions

We built a reproducible pipeline for DRG prediction from Hospital El Pino data. EDA showed obstetric dominance, heavy imbalance, and sparse secondary fields. Sparse encoding plus filtering and class weighting enabled feasible training. Random forest outperformed logistic regression. Frequent DRGs are predicted more reliably; rare DRGs remain difficult, consistent with data limits.

7 Limitations and future work

Limitations include single-hospital scope, residual imbalance among 229 classes, independent binary coding of ICD tokens (no hierarchy), lack of gradient-boosted baselines (e.g., XGBoost, LightGBM), and a random split rather than temporal validation. Future work can explore hierarchical DRG prediction, clinically motivated embeddings, multi-site data, and chronologically split evaluation.

Acknowledgment

The authors thank course instructors and Hospital El Pino for the anonymized teaching dataset.

References

[1] World Health Organization, *International Statistical Classification of Diseases and Related Health Problems, 10th Revision (ICD-10)*, WHO, 2019.

[2] R. F. Averill *et al.*, "Development of the Diagnosis Related Groups," in *Proc. 3rd Annu. Symp. Comput. Appl. Med. Care*, 1989, pp. 475–482.

[3] N. Goldfield *et al.*, "Identifying Potentially Preventable Readmissions," *Health Aff.*, vol. 27, no. 2, pp. 430–436, 2008.

[4] P. Xie and E. P. Xing, "A Neural Architecture for Information Extraction from Clinical Text," in *Proc. Workshop Mach. Learn. Health*, 2018.

[5] J. Mullenbach *et al.*, "Explainable Prediction of Medical Codes from Clinical Text," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2018, pp. 532–539.